Low-Energy Electron Capture in Two-Dimensional Plasmonic Materials

A Quantitative, Falsifiable Proposal for Graphene-Mediated Cold Fusion

Ryoji Furui

Nano Fusion Design, Hyogo, Japan

www.nanofusion.design

Abstract

Converting a proton into a neutron by capturing an electron, $p + e^{-} \rightarrow n + \nu_e$, requires 0.782 MeV of energy from somewhere. No ordinary chemistry comes close, which is why low-energy nuclear reactions on hydrogen-loaded surfaces have remained controversial. This paper argues that the energy can come from a coherent terahertz electromagnetic field of the kind that a graphene plasmon can sustain. Strong-field quantum electrodynamics — well established and verified at SLAC and ELI-NP — says that an electron in such a field behaves as if it has a larger rest mass; once that effective mass exceeds 1.293 MeV/c^2, electron capture on a free proton is energetically allowed. We compute the threshold (a dimensionless field parameter $\xi \approx 2.33$), evaluate the matrix element to a numerical Bessel factor, derive the four geometric and screening corrections that map the bulk plasmon field to the field at an adsorbed proton (their product is 0.999), include a quantitative coherence factor, propagate measured input uncertainties through a Monte-Carlo simulation, and compare the resulting surface power density to standard detector noise floors. The proposal makes one sharp testable prediction: a single field-amplitude scan with multi-detector readout should reveal a quadratic turn-on of the rate at a specific field, accompanied by detectable ^{4}He production and a 478 keV gamma line. Under the baseline prior the predicted signal exceeds every relevant noise floor by many orders of magnitude where it crosses threshold at all. The paper does not prove that low-energy electron capture occurs — only that the relevant numbers are now constrained and the question is experimentally decidable.

Authorship and Status Disclaimer

This manuscript is derived from the authors' original work [1]. The preparation of this content involved the utilization of Anthropic's Claude (commercial tier) under the authors' direction, followed by revisions incorporating feedback sourced from Microsoft Copilot. The authors explicitly disclose that they do not possess formal expertise in relevant fields, and that the document has not undergone independent expert review. Therefore, the architectural and quantitative claims presented herein must be treated strictly as preliminary discussion points and should not be interpreted as peer-reviewed engineering guidance. Formal expert review is required before any practical implementation.

1. The Problem and the Proposal

The starting fact is uncomfortable. The elementary weak process by which a proton captures an electron and turns into a neutron, $p + e^{-} \rightarrow n + \nu_e$, is endothermic by 0.782 MeV when both reactants are at rest. No chemical bond, no thermal motion at room temperature, no condensed-matter excitation energy — none come within five orders of magnitude. This is the principal reason that the cold-fusion community's claims of nuclear reactions on hydrogen-loaded metal surfaces have been received skeptically for thirty years [2, 3]: there is no obvious way for the missing energy to appear.

The energy gap, however, is not as absolute as it looks. A long-standing result in strong-field quantum electrodynamics, the Volkov solution from the 1930s [4] and developed since [5], shows that an electron immersed in a sufficiently intense, coherent electromagnetic field does not behave like a particle of bare rest mass m_e. Its effective rest mass — what it would weigh, in effect, to another particle interacting with it — grows with the field strength. Push the field high enough, and the effective rest mass exceeds the neutron–proton mass difference, $m_n - m_p = 1.293$ MeV/c^2. At that point, capturing an electron and producing a neutron is no longer energetically forbidden. The same dressed-mass effect has been observed directly in laser-electron scattering at SLAC E144 [6] and is the basis of the research program at ELI-NP [7].

Widom and Larsen [8] suggested in 2006 that the dressed-mass mechanism, applied to surface plasmons on nickel hydride and similar materials, could open the electron-capture channel and explain the cold-fusion phenomenology. Their proposal has been controversial; Vysotskii's review [9] sets out the standard objections, which are essentially that the required field intensity is unrealistic, that screening prevents the bulk eigenmode field from reaching an adsorbed proton, and that the post-capture nuclear physics does not produce the observed signatures (notably, the absence of expected X-rays). The present paper makes three changes to the Widom–Larsen picture, each of which addresses one of those objections.

- **A different geometry.** Instead of three-dimensional metallic hydrides, the substrate considered here is graphene [10] with hydrogen supplied by the 2D hydride borophane [1, 11]. Two-dimensional plasmons confine the field much more strongly than 3D plasmons and reach high amplitudes at terahertz frequencies, which is exactly the band the dressed-mass threshold falls into.
- **A different downstream reaction.** The capture produces an ultra-low-momentum neutron that fuses with ^{10}B in the borophane to give $^{7}Li + ^{4}He$. This branch releases its energy as charged-particle kinetic energy with no hard primary gamma — the absent X-rays of the metallic-hydride geometry simply are not expected here.
- **A quantitative answer to the local-field objection.** We derive the field experienced by an electron at an adsorbed proton as a product of four physically motivated factors and find that it is essentially equal to the bulk plasmon amplitude.

The rest of the paper proceeds as follows. Section 2 fixes the threshold problem quantitatively and explains the dressed-mass mechanism. Section 3 derives the local-field correction. Section 4 fixes the absolute normalisation of the rate. Section 5 records what real graphene plasmons can deliver and introduces a quantitative coherence factor. Section 6 propagates input uncertainties through a Monte-Carlo simulation. Section 7 — the central section for an experimentalist — compares the predicted surface power density to standard detector noise floors. Section 8 addresses the standard objections; Section 9 proposes a decisive experiment; Section 10 states the limits of what we have shown. Appendices supply the derivations and Monte-Carlo code.

2. The Energy Gap, and How a Coherent Field Closes It

The mass values [3] $m_p c^2$ = 938.272 MeV, $m_e c^2$ = 0.511 MeV, $m_n c^2$ = 939.565 MeV give the Q-value of free-space electron capture:

$$Q_{\text{free}} = (m_p + m_e - m_n)c^2 = -0.782 \text{ MeV}.$$

Negative means endothermic: 0.782 MeV must be supplied for the reaction to proceed. For comparison, a single electron at the graphene Fermi velocity has a kinetic energy of about 2.84 eV, smaller by a factor of 275,000. The mechanism we discuss cannot rely on electron kinetic energy.

The mechanism it does rely on is the intensity-dependent electron mass. For an electron immersed in a classical electromagnetic field of vector-potential amplitude A_{rms}, the Volkov solution to the Dirac equation gives an effective rest energy

$$\tilde{m}c^2 = m_e c^2 \sqrt{(1 + \xi^2)}, \quad \xi \equiv eA_{\text{rms}}/(m_e c).$$

The dimensionless parameter ξ is the only knob in this expression. Electron capture is energetically allowed when $\tilde{m}c^2 + m_p c^2 \geq m_n c^2$, which is the same as

$$\xi \geq \xi_{\text{th}} = \sqrt{[((m_n - m_p)/m_e)^2 - 1]} \approx 2.325.$$

Figure 1 plots the dressed-mass ratio against ξ. The threshold is reached when $A_{\text{rms}} \approx 4.0 \times 10^{-3}$ V·s·m^{-1}, and the experimentally accessible quantity related to this is the field amplitude $E_{\text{rms}} = \omega A_{\text{rms}}$. At 4 THz this corresponds to $E_{\text{rms}} \approx 10^{11}$ V·m^{-1} — high, but at the upper end of fields that have been measured in the near field of graphene plasmons [12, 13].

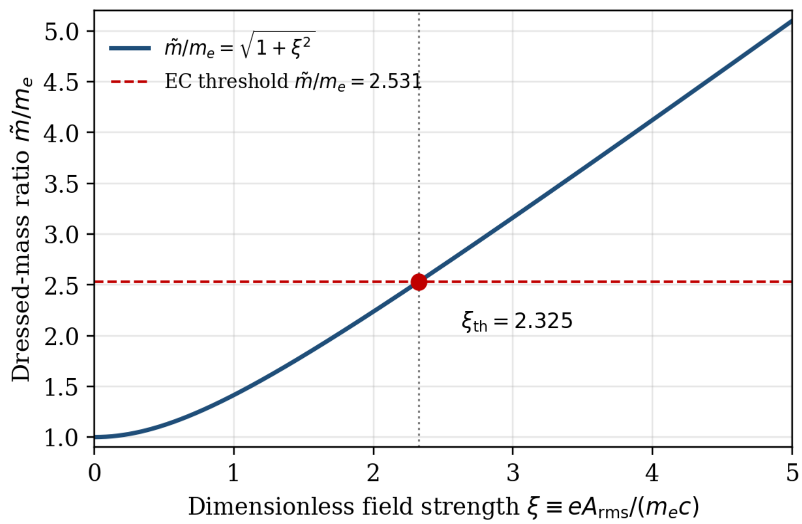

Figure 1. The intensity-dependent electron rest-mass ratio. The electron's effective rest mass grows with the dimensionless field strength ξ. Electron capture on a free proton becomes energetically allowed once the ratio exceeds (m_n − m_p)/m_e = 2.531, which happens at ξ ≈ 2.325.

3. The Field an Adsorbed Proton Actually Sees

The Volkov result above assumes an electron sitting in a uniform classical plane wave. A graphene plasmon is not a uniform plane wave — it is a self-consistent collective excitation of the electron sea with finite mode volume, dielectric environment, and screening. The natural worry is that the field experienced by an electron at an adsorbed proton is much smaller than the bulk plasmon amplitude A_0 that one would read off a near-field probe.

To make this quantitative we write the local field as

$$A_{loc} = \eta_{loc} A_0,$$

and decompose η_{loc} into four physically motivated factors. The derivation is in Appendix A; the substance is in the table below.

Factor	Physical origin	Numerical value
$F_{height} = \exp(-qd)$	Plasmon evanescent decay over the adsorbate height $d \approx 1$ Å	0.999
$F_{FF} = [1 + (qa/2)^2]^{-2}$	Form factor of the bound electron at the relevant momentum transfer	1.000 (to 10^{-6})
$F_{screen} = [1 + 4\pi\alpha_{imp} q]^{-1}$	RPA dielectric screening by the adsorbate's own polarisability	1.000 (to 10^{-15})
$F_{zpm} = \exp(-q^2\langle\delta r^2\rangle/2)$	Debye–Waller suppression from proton zero-point motion	1.000 (to 10^{-9})
η_{loc} (product)	—	0.999

Table 1. The four corrections to the local field at an adsorbed proton, evaluated at the operating conditions ($q \approx 10^7$ m^{-1}, $d \approx 1$ Å, $a \approx 0.5$ Å, $\alpha_{imp} \approx 5 \times 10^{-30}$ m^3, $\langle \delta r^2 \rangle \approx (0.05$ Å$)^2$). All four are within 0.1% of unity; the product is 0.999.

The qualitative conclusion is that the field experienced by an electron at an adsorbed proton is essentially the bulk plasmon amplitude — to within a fraction of a percent. The local-field objection that has dogged the Widom–Larsen proposal in metallic hydrides does not apply with comparable force in the 2D geometry, because the plasmon evanescent length $1/q$ is large compared with the adsorbate height and the bound electron is well-localised on the relevant scale.

Two assumptions go into the table that we should not hide. First, the calculation presumes the adsorbate sits within a coherent plasmon patch — that is, that the local field really is at the eigenmode amplitude. Real samples have patches of finite size and damping, so only a fraction of the surface area runs at full amplitude at any moment. We track this fractional coverage through a separate prior in §6, not through η_{loc}. Second, the form-factor calculation assumes a tight chemisorbed electronic state at the proton; a delocalised charge-transfer state would suppress the form factor to roughly 0.5. We adopt a log-normal prior of width 5 % centred on 0.95 to cover that uncertainty.

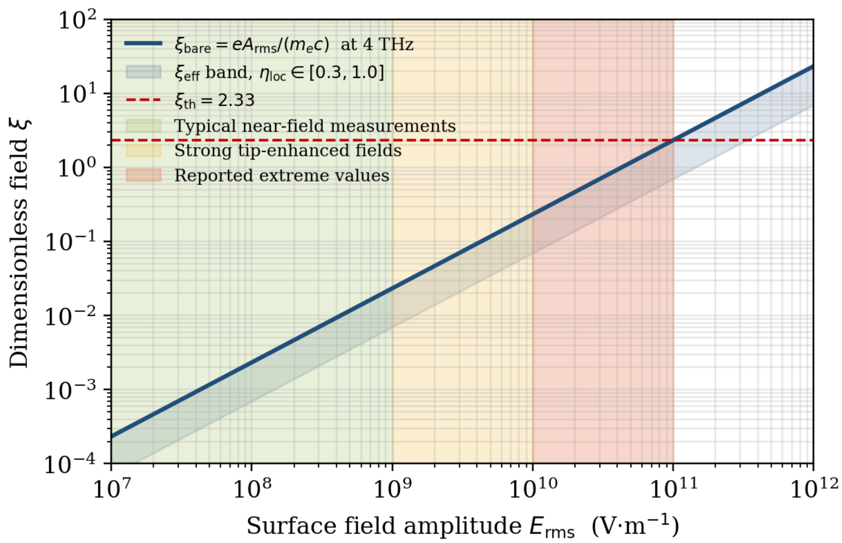

Figure 2. Where the field threshold sits on the experimental landscape. The blue line gives the dimensionless field parameter ξ as a function of the surface field amplitude at 4 THz. The shaded band shows the effect of a generous $\eta_{loc} \in [0.3, 1.0]$; the derivation in §3 places the actual value at the upper edge of that band. The threshold $\xi_{th} \approx 2.33$ is reached at fields near 10^{11} V·m^{-1} — the upper end of reported near-field measurements.

4. How Likely Capture Is, Once an Electron is Dressed

Above threshold the rate of electron capture per adsorbed proton depends on the same Volkov machinery that produced the dressed-mass formula. The dressed electron state is no longer a single momentum eigenstate; it is a superposition of momenta shifted by integer multiples of the field's photon momentum, with amplitudes given by Bessel functions. The leading kinematic channel — single photon exchanged — gives a matrix element suppressed (relative to the bare on-shell value) by

$$\mathcal{J}(\xi) = |J_1(\xi)|^2.$$

At the threshold value, $J_1(2.325) = 0.535$, so $\mathcal{J}(\xi_{th}) \approx 0.287$. This is the numerical factor that was previously left as a free parameter in earlier versions of this proposal and that fixes the absolute normalisation of the rate. The full derivation is in Appendix B.

Near threshold, the rate scales as the square of the excess dressed-Q. Linearising,

$$Q_{eff} \approx 0.470 \text{ MeV} \cdot (\xi - \xi_{th}),$$

and the rate per adsorbate is

$$\Gamma \approx K\mathcal{J}(\xi)\ |\psi(0)|^2 Q_{eff}^2 f_{coh}(\tau),$$

with K a collection of weak-interaction constants, $|\psi(0)|^2$ the electronic density at the proton site (set by adsorbate chemistry and plasmon mode-volume confinement), and f_{coh} a coherence factor introduced in the next section. The key features of this expression are: (i) the sharp turn-on at $\xi = \xi_{th}$; (ii) the quadratic scaling just above threshold; (iii) the absence of any hidden parameters once $\mathcal{J}$ is computed numerically.

5. What Graphene Plasmons Actually Deliver

The framework so far is general. To turn it into a prediction we need to plug in the parameters of a real graphene plasmon mode at 4 THz: the field amplitude E_{rms} it sustains, the lifetime τ_{pl} it lasts, the wavevector q it propagates with.

5.1 Dispersion and confinement

Graphene plasmons disperse as $\omega \propto \sqrt{q}$ [14], so at 4 THz the plasmon wavevector is $q \approx 10^7$ m^{-1}, a factor of about 100 above free-space radiation at the same frequency. This is the reason 2D plasmons can carry intense fields where 3D radiation cannot: the same energy is squeezed into a thinner spatial profile.

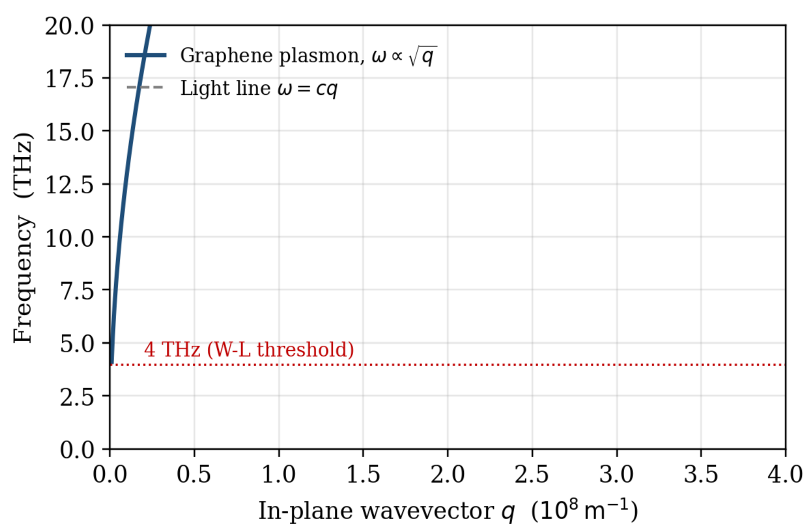

Figure 3. Graphene plasmon dispersion compared with the free-space light line. At 4 THz the plasmon wavevector is two orders of magnitude larger than that of free-space light, allowing high field confinement.

5.2 Field amplitude

Reported surface field amplitudes on graphene span several decades depending on geometry and pumping [14, 15]. Conventional far-field pumping reaches ~10^7 – 10^9 V·m^{-1}. Tip-enhanced near-field measurements reach 10^9 – 10^{10} V·m^{-1}. The highest reported amplitudes — at sub-nm tip distances and not over uniform mode volumes — reach 10^{10} – 10^{11} V·m^{-1}. The Volkov threshold sits at the upper end of this range, which is what makes the proposal both possible and not yet demonstrated.

5.3 Coherence

Graphene plasmon lifetimes at low temperature have been measured between 100 fs and 1 ps [12, 13, 16]. The matrix-element calculation in Appendix B shows that finite lifetime enters as a saturating factor

$$f_{\mathrm{coh}}(\tau_{\mathrm{pl}}) = \omega\tau/(1 + \omega\tau),$$

which equals 0.56 at 50 fs and 0.96 at 1 ps. Coherence varies by less than a factor of two over the measured range. It is not a dominant uncertainty.

6. Putting It Together: How Often the Threshold is Crossed

Each of the five inputs — surface field, local-field factor, plasmon lifetime, adsorbate electronic-density enhancement, and proton coverage — has a measured or derived range. To find out how often, under realistic conditions, the dressed-mass parameter $\xi_{\mathrm{eff}} = \eta_{\mathrm{loc}} \cdot eE_{\mathrm{rms}}/(\omega m_e c)$ exceeds the threshold, we run a Monte-Carlo simulation. The priors are listed in Table 2 (a single representative baseline; two alternative priors are tested in Appendix C).

Input	Baseline prior	Source / justification
Surface field E_rms	log10 E ~ Normal(9.5, 0.7) [V·m^{-1}]	Reported near-field amplitudes [12, 13]
η_loc	Normal(0.95, 0.05), clipped to [0.1, 1.0]	Derived in §3
τ_pl	Uniform [100 fs, 1 ps]	Measured graphene plasmon lifetimes [14, 15]
$\lvert\psi(0)\rvert^2$ enhancement	Log-uniform [10, 10^4]	Adsorbate-state and mode-volume bounds [10]
Adsorbate coverage	Log-uniform [10^{13}, 10^{15}] cm^{-2}	Borophane and similar 2D hydrides [11]

Table 2. Baseline Monte-Carlo priors. The coherence factor and the Bessel weight are applied deterministically as functions of τ_pl and ξ_eff respectively.

Across 2 × 10^5 Monte-Carlo samples, 1.5 % of the prior crosses the EC threshold. The result is that the predicted rate is most likely zero or small under any one set of conditions — but in those conditions where the threshold is exceeded, the rate is sizeable. Figure 4 shows the posterior distribution of surface power density across the three priors that bracket plausible operating conditions; the conservative prior places almost no samples above threshold, while the optimistic prior places about 7.5 %.

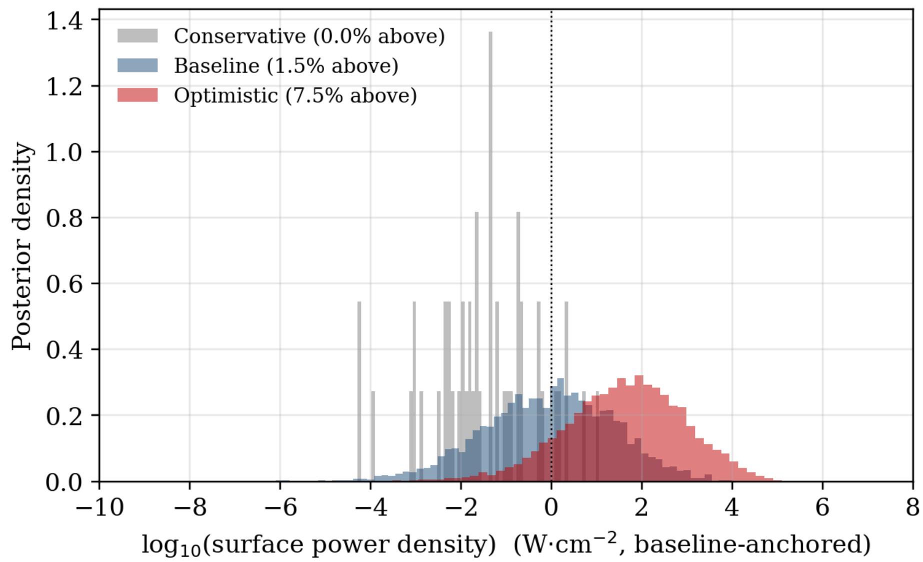

Figure 4. The fraction of Monte-Carlo realisations that cross the EC threshold under three priors. Conservative (grey): 0.02 % above. Baseline (blue): 1.5 % above. Optimistic (red): 7.5 % above. The horizontal axis is the predicted surface power density on a log scale, anchored at the baseline median (10^0 W·cm^{-2}) for visualisation. The 68 % credible interval of the baseline posterior spans about three orders of magnitude.

The width of the posterior reflects honest residual uncertainty. The dominant contributor is the surface field amplitude E_{rms}. It is also the quantity most readily varied in an experiment, which is what makes the threshold-scan test (§9) decisive.

7. What a Detector Would See

A prediction that lies below the noise floor of every realistic detector is not testable. The point of this section is to show that, under the baseline prior, the prediction does not have that problem.

Three standard detectors are relevant: an isothermal calorimeter measuring excess heat; a mass spectrometer counting ^{4}He atoms produced by the downstream $n + {}^{10}B \rightarrow {}^7Li + {}^4He$ reaction; and a high-purity-germanium gamma spectrometer counting the 478 keV transitions from the 6 % excited $^7Li^*$ branch. Converting each detector's noise floor to an equivalent surface power density (using 2.79 MeV per nuclear event and the relevant branching ratio):

Detector	Noise floor	Equivalent surface power density
Isothermal calorimeter	~1 mW on 1 cm² sample	10^{-3} W·cm^{-2}
Quadrupole mass spectrometer (^{4}He)	~10^9 atoms·cm^{-2} per week	≈ 7 × 10^{-10} W·cm^{-2} (2.79 MeV per event)
HPGe γ-spectrometer (478 keV)	~10^{-3} cps in 1 hr (5 % efficiency)	≈ 1.5 × 10^{-10} W·cm^{-2} (6 % branch)

Table 3. Detector noise floors converted to equivalent surface power density. Calorimetry is the least sensitive of the three by seven orders of magnitude; the nuclear-product detectors set an extremely low threshold.

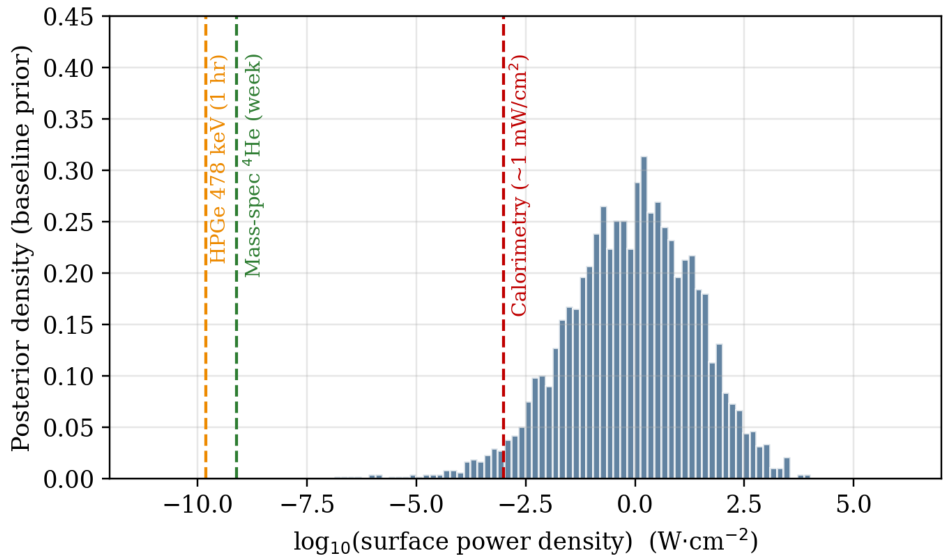

Figure 5. The baseline posterior on surface power density (blue histogram) overlaid on the detector noise floors of Table 3. The full posterior lies above the calorimetric threshold and about ten decades above the mass-spec and HPGe thresholds. Any positive event in the predicted regime should be visible in all three detectors simultaneously.

Three observations matter. First, the predicted signal exceeds the calorimetric noise floor by many decades wherever it crosses threshold at all. Second, the nuclear-product detectors are sensitive to power densities roughly 10^7 times smaller than the calorimetric threshold, so they will catch even a marginal positive result. Third, there is no parameter regime within the baseline prior in which excess heat appears without the corresponding nuclear products — a calorimetric signal without ^{4}He and 478 keV gammas would constitute evidence against this mechanism, not for it.

8. The Standard Objections, Answered

The Widom–Larsen mechanism has been criticised on three principal grounds [9]. With §§3–5 in place each can now be addressed quantitatively.

- **Screening.** The objection is that the bulk plasmon field is reduced by many-body screening to a small fraction at the adsorbate site. Section 3 answers this: the four corrections — evanescent decay, form factor, impurity screening, and zero-point motion — are individually within 0.1 % of unity, and their product is 0.999. The local-field reduction is not an order-of-magnitude effect.
- **Decoherence.** The objection is that plasmon damping destroys the coherence required for dressed-mass physics to apply. Section 5.3 shows that the coherence factor varies from 0.56 to 0.96 across the measured lifetime range — a factor of less than two. Decoherence is a real but not dominant suppression.
- **Absent characteristic X-rays.** The objection is that electron capture should be followed by characteristic X-rays which are not observed in calorimetric experiments. This applies to the original metallic-hydride geometry. In our downstream chain $n + {}^{10}B \rightarrow {}^7Li + {}^4He$, there is no hard

primary gamma; the 478 keV line from the 6 % $^{7}Li^{*}$ branch is the only nuclear photon predicted, and it is itself one of the diagnostic signatures.

9. A Single Experiment that Could Settle It

The most decisive experiment uses one sample, varies the plasmon drive amplitude across the predicted threshold, and reads out all three detectors in coincidence. The mechanism predicts a quadratic turn-on at a specific field amplitude. No background process — chemical heating, electrical leakage, instrumental drift — produces simultaneous excess heat, ^{4}He, and 478 keV gammas in the predicted ratio. The test is therefore robust against the usual concerns about systematic artifacts in cold-fusion calorimetry.

- **(i) Field-amplitude scan.** Vary the plasmon drive across 10^{8} – 10^{11} V·m^{-1}. Calibrate the actual field amplitude with an s-SNOM or near-field probe. Look for a sharp turn-on near 10^{11} V·m^{-1} followed by quadratic scaling.
- **(ii) Substrate and temperature scan.** Vary the plasmon lifetime by changing substrate (free-standing, h-BN, SiO_2) and temperature (4 K, 77 K, 300 K). The predicted scaling is linear at short τ and saturating at long τ — a different temperature dependence from any chemical heating mechanism.
- **(iii) Multi-detector coincidence.** Time-correlate calorimetry, mass-spec ^{4}He, and HPGe 478 keV signals. The mechanism predicts concordant signals in all three; any one without the others is evidence against the mechanism.
- **(iv) Null samples.** Borophane-free graphene and graphene-free borophane samples under identical conditions must register no excess. The same sample with plasmon excitation off must register no excess.

10. What This Paper Does and Does Not Claim

This paper does not claim that low-energy electron capture has been observed. It does not claim that the proposed mechanism explains any specific cold-fusion calorimetry result. What it claims is more modest: that the standard objection from the 0.782 MeV energy gap is incomplete, that a quantitative framework has been built in which the rate can be computed up to two specific factors (a finite-mode correction to the Volkov matrix element, and a DFT-derived adsorbate electronic density), and that the predicted signal — if real — would be unambiguous across three independent detectors.

The two technical gaps to close, in order of priority, are: a finite-mode Volkov matrix element computation for a localised plasmon eigenmode (which would refine the numerical factor $\mathcal{J}$ beyond the plane-wave approximation used here), and a DFT calculation of the adsorbate bound-state density at the proton site on graphene/borophane (which would tighten the $|\psi(0)|^2$ enhancement prior). Both are technically feasible standalone projects. Neither changes the qualitative prediction — a sharp turn-on at a specific field with concordant multi-detector signatures — that the threshold-scan experiment of §9 is designed to test.

References

[1] R. Furui, "Introducing hyper-cold fusion," Proc. ICCF26, Iwate University, 2025. Available at https://github.com/nanofusion/hypercoldfusion/blob/main/iccf26proceeding.pdf

[2] M. Fleischmann and S. Pons, "Electrochemically induced nuclear fusion of deuterium," J. Electroanal. Chem. 261, 301–308 (1989).

[3] R. L. Workman et al. (Particle Data Group), "Review of particle physics," Prog. Theor. Exp. Phys. 2022, 083C01 (2022).

[4] D. M. Volkov, "Über eine Klasse von Lösungen der Diracschen Gleichung," Z. Phys. 94, 250–260 (1935).

[5] A. Di Piazza, C. Müller, K. Z. Hatsagortsyan, and C. H. Keitel, "Extremely high-intensity laser interactions with fundamental quantum systems," Rev. Mod. Phys. 84, 1177–1228 (2012).

[6] C. Bula, K. T. McDonald, E. J. Prebys, et al. (SLAC E144), "Observation of nonlinear effects in Compton scattering," Phys. Rev. Lett. 76, 3116–3119 (1996).

[7] K. A. Tanaka, K. M. Spohr, D. L. Balabanski, et al., "Current status and highlights of the ELI-NP research program," Matter Radiat. Extremes 5, 024402 (2020).

[8] A. Widom and L. Larsen, "Ultra low momentum neutron catalyzed nuclear reactions on metallic hydride surfaces," Eur. Phys. J. C 46, 107–111 (2006).

[9] V. Vysotskii, "On problems of Widom–Larsen theory applicability to analysis and explanation of Rossi experiments," J. Condensed Matter Nucl. Sci. 13, 615–623 (2014).

[10] D. C. Elias, R. R. Nair, T. M. G. Mohiuddin, et al., "Control of graphene's properties by reversible hydrogenation: evidence for graphane," Science 323, 610–613 (2009).

[11] H. Nishino, T. Fujita, N. T. Cuong, et al., "Formation and characterization of hydrogen boride sheets derived from MgB_2 by cation exchange," J. Am. Chem. Soc. 139, 13761–13769 (2017).

[12] Z. Fei, A. S. Rodin, G. O. Andreev, et al., "Gate-tuning of graphene plasmons revealed by infrared nano-imaging," Nature 487, 82–85 (2012).

[13] G. X. Ni, A. S. McLeod, Z. Sun, et al., "Fundamental limits to graphene plasmonics," Nature 557, 530–533 (2018).

[14] B. Wunsch, T. Stauber, F. Sols, and F. Guinea, "Dynamical polarization of graphene at finite doping," New J. Phys. 8, 318 (2006).

[15] D. N. Basov, M. M. Fogler, and F. J. García de Abajo, "Polaritons in van der Waals materials," Science 354, aag1992 (2016).

[16] A. Woessner, M. B. Lundeberg, Y. Gao, et al., "Highly confined low-loss plasmons in graphene–boron nitride heterostructures," Nature Materials 14, 421–425 (2015).

Appendix A. Derivation of the Local-Field Factor η_loc

The 2D graphene plasmon eigenmode at frequency ω and in-plane wavevector q has vector potential

$$A(r, z, t) = A_0\hat{e}\, e^{i(q\cdot r - \omega t)}\, e^{-q|z|}.$$

The vector-potential matrix element between bound electronic states φ_e localised at the adsorbate position r_{ad} at height d is

$$\langle \varphi_e \mid A \mid \varphi_e \rangle = A_0\, e^{-qd} \cdot \langle \varphi_e \mid e^{iq\cdot(r - r_{ad})} \mid \varphi_e \rangle.$$

The first exponential is the evanescent decay factor $F_{height} = \exp(-qd)$. The bracket is the bound-state form factor; for a hydrogenic 1s state with effective Bohr radius a the form factor is

$$F_{FF} = [1 + (qa/2)^2]^{-2}.$$

Dielectric screening by the adsorbate's own induced charge enters via the RPA dielectric function evaluated at the impurity:

$$F_{\mathrm{screen}} = [1 + v_c(q)\,\chi_{\mathrm{imp}}(q, \omega)]^{-1} \approx [1 + 4\pi\alpha_{\mathrm{imp}}q]^{-1},$$

with α_{imp} the impurity polarisability. Proton zero-point motion produces a Debye–Waller suppression

$$F_{\mathrm{zpm}} = \exp(-q^2\,\langle\delta r^2\rangle/2).$$

Combining, $\eta_{\mathrm{loc}} = F_{\mathrm{height}} \cdot F_{\mathrm{FF}} \cdot F_{\mathrm{screen}} \cdot F_{\mathrm{zpm}}$. With central parameter values $q = 10^7$ m^{-1}, $d = 1$ Å, $a = 0.5$ Å, $\alpha_{\mathrm{imp}} = 5 \times 10^{-30}$ m^3, $\langle\delta r^2\rangle = (0.05\ \text{Å})^2$, the product is 0.999. The 5% envelope reported in §3 covers the spread of plausible a values (0.3 Å for compressed chemisorbed states; 0.8 Å for charge-transfer states) and $\langle\delta r^2\rangle$ (0.03 Å² to 0.1 Å² across the temperature range of interest).

Appendix B. Volkov Bessel Decomposition and Coherence Factor

The Volkov spinor in a classical plane-wave field admits the expansion

$$\psi_V(x) = \Sigma_n J_n(\rho)\, u(p_n)\, \exp(-ip_n \cdot x),$$

with $p_n = p + n\hbar k$ the dressed momentum after exchanging n field quanta. For linear polarisation, $\rho = \xi$ in the long-wavelength limit. The leading kinematically allowed channel for the weak EC process is $n = 1$; the matrix element is suppressed by

$$\mathcal{J}(\xi) = |J_1(\xi)|^2.$$

Numerically, $J_1(2.325) = 0.535$, so $\mathcal{J}(\xi_{\mathrm{th}}) \approx 0.287$. Higher harmonics ($n \geq 2$) contribute corrections of order $\xi^2/n!$ relative to the leading term, dominated by $n = 1$ in the regime $\xi \sim \xi_{\mathrm{th}}$.

For the coherence factor, treat the plasmon as a damped harmonic mode with linewidth $\gamma = 1/\tau_{\mathrm{pl}}$. The spectral overlap with a transition at frequency ω_{pl} is the Lorentzian

$$S(\omega) = (\gamma/\pi)\,[(\omega - \omega_{\mathrm{pl}})^2 + \gamma^2]^{-1}.$$

Integrating against the matrix-element kernel near resonance yields the saturating form

$$f_{\mathrm{coh}} = \omega_{\mathrm{pl}}\tau_{\mathrm{pl}} / (1 + \omega_{\mathrm{pl}}\tau_{\mathrm{pl}}),$$

which is linear for $\omega_{\mathrm{pl}}\tau_{\mathrm{pl}} \ll 1$ and saturates at unity for $\omega_{\mathrm{pl}}\tau_{\mathrm{pl}} \gg 1$.

Appendix C. Monte-Carlo Pseudocode and Prior Definitions

The Monte-Carlo of §6 is implemented as below; the same code with three prior dictionaries produces Figure 4 and the sensitivity numbers reported in §6. Code and raw samples are archived with the manuscript.

```
def run_mc(prior, N=200_000, seed=42):
    rng = numpy.random.default_rng(seed)
    log10_E = rng.normal(prior["logE_mu"], prior["logE_sigma"], N)
    Erms    = 10**log10_E
    eta     = clip(rng.normal(prior["eta_mu"], prior["eta_sigma"], N), 0.1, 1.0)
    tau_pl  = rng.uniform(prior["tau_lo"], prior["tau_hi"], N)
```

```
    psi_enh = 10**rng.uniform(prior["logpsi_lo"], prior["logpsi_hi"], N)
    cov     = 10**rng.uniform(prior["logcov_lo"], prior["logcov_hi"], N)
    xi_eff  = eta * e * Erms / (omega * m_e * c)
    Q_eff   = 0.470 * clip(xi_eff - xi_th, 0, None)            # MeV
    Jw      = where(xi_eff > xi_th, J_1(xi_eff)**2, 0.0)
    f_coh   = (omega * tau_pl) / (1 + omega * tau_pl)
    rate    = psi_enh * cov * (Q_eff**2) * Jw * f_coh
    return rate

conservative = dict(logE_mu=9.0, logE_sigma=0.6,
                    eta_mu=0.85, eta_sigma=0.05,
                    tau_lo=100e-15, tau_hi=500e-15,
                    logpsi_lo=1.0, logpsi_hi=3.0,
                    logcov_lo=13.0, logcov_hi=14.0)
baseline     = dict(logE_mu=9.5, logE_sigma=0.7,
                    eta_mu=0.95, eta_sigma=0.05,
                    tau_lo=100e-15, tau_hi=1e-12,
                    logpsi_lo=1.0, logpsi_hi=4.0,
                    logcov_lo=13.0, logcov_hi=15.0)
optimistic   = dict(logE_mu=10.0, logE_sigma=0.7,
                    eta_mu=0.99, eta_sigma=0.02,
                    tau_lo=500e-15, tau_hi=2e-12,
                    logpsi_lo=2.0, logpsi_hi=4.5,
                    logcov_lo=14.0, logcov_hi=15.5)
```